Experiment 1: ER Diagram for Indian E-Commerce Platform

1. Title

Design an ER Diagram for an Indian E-Commerce Platform

2. Aim

To design an Entity-Relationship (ER) diagram for an Indian e-commerce platform that represents customers, orders, products, sellers, payments, deliveries, addresses, and categories along with their attributes and relationships.

3. Theory

An Entity-Relationship (ER) model is a conceptual model used to represent the structure of a database. It shows the important entities of a system, their attributes, and the relationships between them.

In an e-commerce system, entities such as **Customer, Order, Product, Seller, Payment, Delivery, Address, and Category** are required to store and manage different types of information.

The ER diagram uses different components:

- **Entity:** Represents a real-world object such as Customer, Product, or Order.
- **Attribute:** Represents a property of an entity, such as customer name or product price.
- **Primary Key (PK):** Uniquely identifies each entity record, such as customer_id or product_id.
- **Composite Key:** A key formed using more than one attribute. In the diagram, ORDER_ITEM uses (order_id, product_id) as a composite key.
- **Multi-valued Attribute:** An attribute that can contain multiple values, such as phone numbers.
- **Composite Attribute:** An attribute that can be divided into smaller attributes. For example, Address can contain house number, street, city, state, pincode, and country.
- **Relationship:** Represents an association between entities, such as Customer places Order.
- **Cardinality:** Defines the number of entities participating in a relationship, such as 1:N or 1:1.

- **Participation Constraint:** Defines whether participation of an entity in a relationship is mandatory or optional.
- **Specialization (ISA):** Divides a general entity into more specific entities. In this diagram, PRODUCT is specialized into PHYSICAL_PRODUCT and DIGITAL_PRODUCT.

DIAGRAM:

